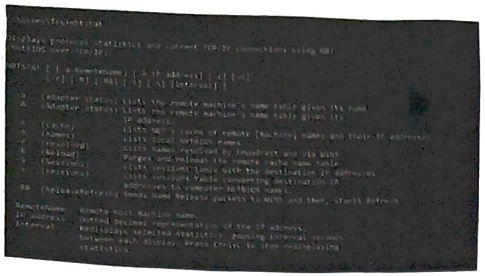
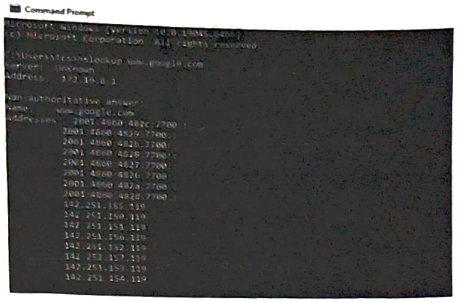
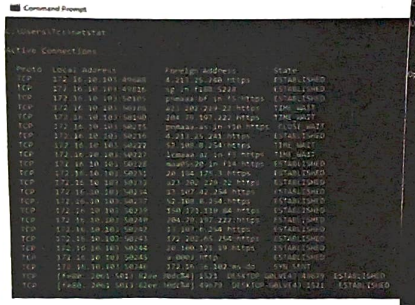
4) nbtstat



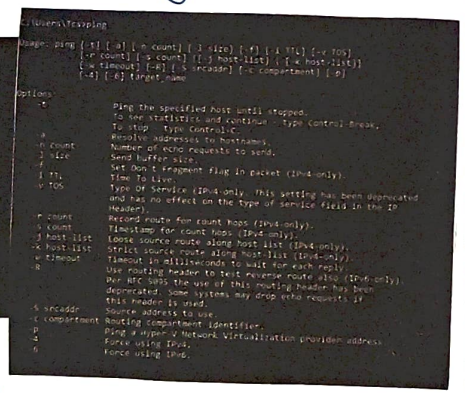
5) nslookup



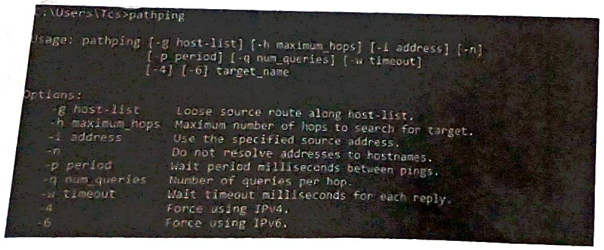
6) net stat



7) ping



8) pathping



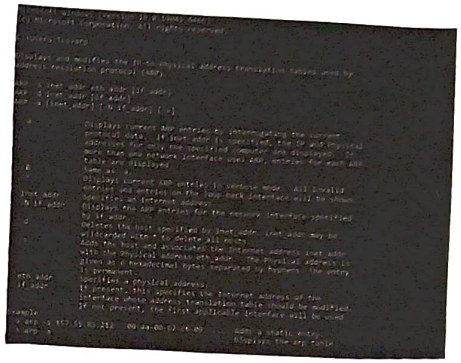
EXP NO - 1
DATE -

STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS

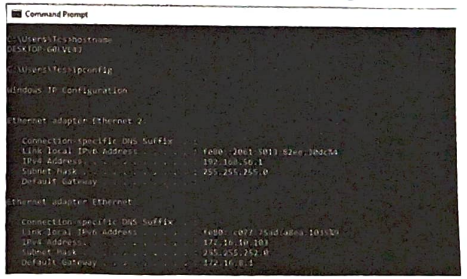
AIM: To study various network commands used in Linux and Windows.

NETWORK COMMANDS IN WINDOWS:

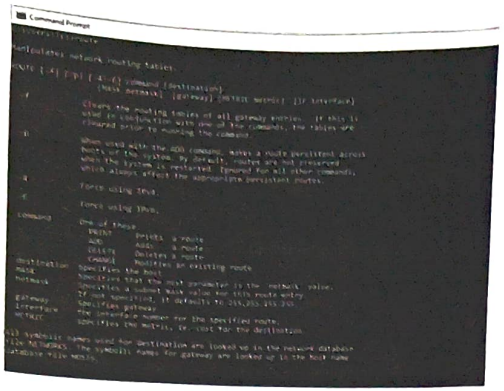
1) arp - Address Resolution Protocol



2) hostname, ipconfig

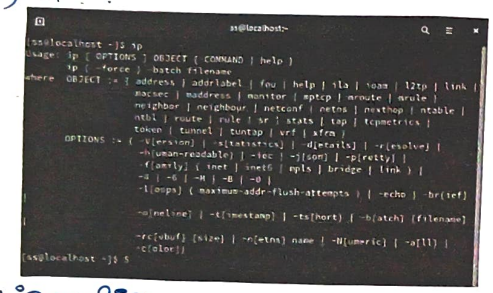


3) route:

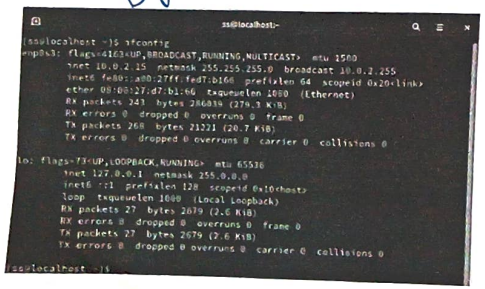


NETWORK COMMANDS IN LINUX:

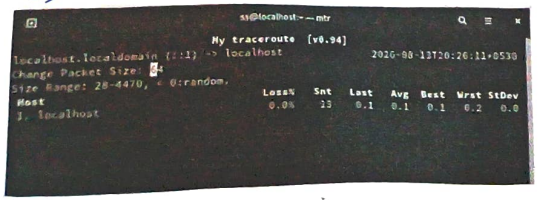
1) ip



2) ipconfig



3) mtr



BASIC NETWORKING COMMANDS:

1) arp-a :- arp is a short form of address resolution protocol, It will show the IP address of your computer along with IP and MAC address of your router.

2) hostname: This command is the simplest of all TCP/IP commands. It simply displays the name of your computer.

3) ipconfig /all :- This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of Ethernet adapter in your system.

4) nbstat-a: This command helps save problems with net BIOS name resolution. (Not stands for Net BIOS over TCP/IP).

5) netstat-(network statistics): netstat displays a variety of statistics about a computers active TCP/IP connections. It is a command line tool for monitoring network connections. both incoming and outgoing as well as viewing routing tables, interface statistics etc..

6) nslookup-(name server lookup) is a tool used to perform DNS lookups in Linux. It is used to display DNS details such as IPaddress of a particular computer. nslookup can operate in two modes: interactive and non-interactive.
eg: nslookup www.google.com

7) pathping: Pathping is unique to windows and is basically a condition of the ping and Tracert commands. Pathping traces the routes to the destination address then launches a 25 seconds test of each router, gather statistics on the rate of data loss along each hop

8) ping (packet Internet Grouper): This command is the best way to test connectivity between two nodes. ping use ICMP (Internet control Message Protocol) to communicate to other devices.

1. #ping hostname (ping localhost)
2. #ping ip address (ping 4.2.2.2)

9) Route :- Route command is used to show manipulate the IP routing table. It is primarily used to setup static routers to specific host or network via an interface.

LINUX NETWORKING COMMANDS:

1) ip: The ip command is one of the basic commands every administrator will need in daily work, form setting up new systems and assigning IP's to trouble shooting systems. The IP command can show address information, manipulate routing, display network various device, interface and tunnels.

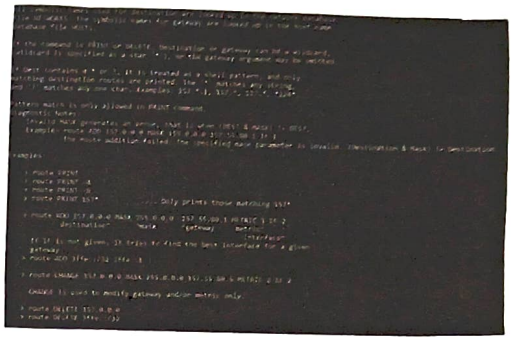
2) ipconfig: The ipconfig command was a staple in many sys admin's tool belt for configuring and trouble shooting networks. It's been replaced by the ip command.

3) mtr: MTR is a program with a command line interface that serves as a network diagnostic and trouble shooting tool combines the functionality of ping and trace route commands.

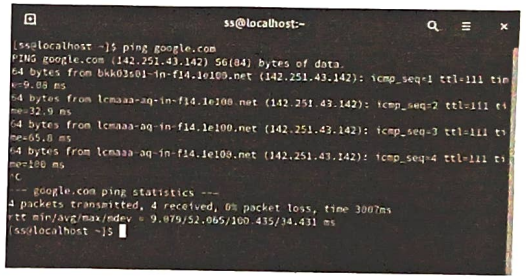
4) tcp dump: The tcp dump command is designed for capturing and displaying packets.

5) ping: ping is a tool that verifies IP level connectivity to another TCP/IP computer by sending Internet control Message Protocol (ICMP) Echo Request message. Ping is the primary TCP/IP command used to troubleshoot connectivity, reachability and name resolution.

4) tcp dump:



5) ping



RESULT: Thus, the Linux and Windows networking commands has been executed successfully.